Lecture Scribes - Lecture Scribe Transformation

L11_S1_A.pdf
PDF

Provide me a lecture scribe for the given lecture. The lecture scribe should be exam-oriented, covering all the details properly. It should only contain the given context, i.e. the contents in the given file. Ensure to include the following:
all the definitions, notations, assumptions, conditions that are given.
theorems, axioms, result statements, proofs and derivations. Each of them should be step by step.
When including the given details, ensure to provide the derivations and proofs of only those things that are covered in the pdf. Also include the worked up examples given in the PDF. Ensure to include the hand written notes and the explanation given to those notes
Note: there should be no content other than the things covered in the given PDF. make sure to not to introduce any new explanations, examples, intuitions, examples, or even a single thing beyond the provided PDF and context.
Make sure to give the result in a properly mannered and structured way, there should be not random things popping up here and there. Make proper sections, subsections and everything should be properly structured in a single shot. Do not include any extra for feedback or "Here is your....", provide only and only the lecture scribe material.

CSE400 – Fundamentals of Probability in Computing

Lecture 11: Transformation of Random Variables

Date: February 10, 2026

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1. Transformation of a Random Variable

1.1 Problem Setting

Let:

- X be a continuous random variable (CRV).
- The PDF $f_X(x)$ of X is known *a priori*.
- Define a new random variable:

$$Y = g(X)$$

- Objective: Find

$$f_Y(y), \quad F_Y(y)$$

(As illustrated in the handwritten notes on page 1 and page 3 .)

1.2 Case 1: $g(\cdot)$ Monotonically Increasing

Assume:

- $Y = g(X)$
- $g(\cdot)$ is **monotonically increasing**
- Hence inverse exists:

$$X = g^{-1}(Y)$$

Step 1: CDF of Y

$$F_Y(y) = \Pr(Y \leqslant y)$$

Substitute $Y = g(X)$:

$$F_Y(y) = \Pr(g(X) \leqslant y)$$

Since $g(\cdot)$ is monotonically increasing:

$$g(X) \leqslant y \Leftrightarrow X \leqslant g^{-1}(y)$$

Therefore,

$$F_Y(y) = \Pr(X \leqslant g^{-1}(y))$$

$$F_Y(y) = F_X(g^{-1}(y))$$

Step 2: Differentiate to Obtain PDF

$$f_Y(y) = \frac{d}{dy} F_Y(y)$$

$$f_Y(y) = \frac{d}{dy} [F_X(g^{-1}(y))]$$

Using chain rule:

$$f_Y(y) = f_X(g^{-1}(y)) \cdot \frac{d}{dy} [g^{-1}(y)]$$

Equivalently (as written on page 3):

$$f_Y(y) = f_X(x) \left| \frac{dx}{dy} \right|_{x=g^{-1}(y)}$$

For increasing transformation:

$$\frac{dx}{dy} > 0$$

Thus,

$$f_Y(y) = f_X(g^{-1}(y)) \cdot \frac{dx}{dy}$$

1.3 Case 2: $g(\cdot)$ Monotonically Decreasing

Assume:

- $g(\cdot)$ is monotonically decreasing.

Step 1: CDF

$$F_Y(y) = \Pr(Y \leqslant y)$$

$$= \Pr(g(X) \leqslant y)$$

Since $g(\cdot)$ is decreasing:

$$g(X) \leqslant y \Leftrightarrow X \geqslant g^{-1}(y)$$

Therefore,

$$F_Y(y) = \Pr(X \geqslant g^{-1}(y))$$
$$= 1 - F_X(g^{-1}(y))$$

(Shown on page 4 .)

Step 2: Differentiate

$$f_Y(y) = \frac{d}{dy}\left[1 - F_X(g^{-1}(y))\right]$$
$$= -f_X(g^{-1}(y)) \cdot \frac{d}{dy}[g^{-1}(y)]$$

Since for decreasing transformation:

$$\frac{dx}{dy} < 0$$

We write:

$$f_Y(y) = f_X(g^{-1}(y)) \left|\frac{dx}{dy}\right|$$

1.4 General Formula

$$\boxed{f_Y(y) = f_X(g^{-1}(y)) \left|\frac{dx}{dy}\right|}$$

with support determined by transformation limits.

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2. Worked Example

(From page 5 .)

2.1 Given

$$X \sim \text{Uniform}(-1, 1)$$

$$f_X(x) = \begin{cases} \frac{1}{2}, & -1 < x < 1 \\ 0, & \text{otherwise} \end{cases}$$

Define:

$$Y = \sin\left(\frac{\pi X}{2}\right)$$

Find: $f_Y(y)$

2.2 Inverse Transformation

$$y = \sin\left(\frac{\pi x}{2}\right)$$

$$x = \frac{2}{\pi}\sin^{-1}(y)$$

2.3 Compute Derivative

$$\frac{dx}{dy} = \frac{2}{\pi} \cdot \frac{1}{\sqrt{1-y^2}}$$

2.4 Apply Formula

$$f_Y(y) = f_X(x)\left|\frac{dx}{dy}\right|$$

Since $f_X(x) = \frac{1}{2}$ in support:

$$f_Y(y) = \frac{1}{2} \cdot \frac{2}{\pi} \cdot \frac{1}{\sqrt{1-y^2}}$$

$$\boxed{f_Y(y) = \frac{1}{\pi\sqrt{1-y^2}}} \quad -1 < y < 1$$

Otherwise 0.

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3. Function of Two Random Variables

(From pages 6–7 .)

3.1 Definition

Let:

$$Z = X + Y$$

Find:

1. $f_Z(z)$
2. $f_Z(z)$, if X and Y are independent
3. If $X \sim N(0,1)$, $Y \sim N(0,1)$, independent, prove:

$$Z \sim N(0,2)$$

4. If X, Y exponential with parameter λ, find $f_Z(z)$

3.2 Derivation Using CDF

$$F_Z(z) = \Pr(Z \leqslant z) = \Pr(X + Y \leqslant z)$$

From the geometric interpretation (page 7):

Region:

$$x + y \leqslant z$$

Therefore:

$$F_Z(z) = \int_{-\infty}^{\infty} \int_{-\infty}^{z-y} f_{XY}(x,y)\, dx\, dy$$

Equivalently changing order:

$$F_Z(z) = \int_{-\infty}^{\infty} \int_{-\infty}^{z-x} f_{XY}(x,y)\, dy\, dx$$

3.3 PDF by Differentiation

$$f_Z(z) = \frac{d}{dz} F_Z(z)$$

When X and Y are independent:

$$f_{XY}(x, y) = f_X(x) f_Y(y)$$

Hence convolution form:

$$\boxed{f_Z(z) = \int_{-\infty}^{\infty} f_X(x) f_Y(z - x)\, dx}$$

3.4 Normal Case

Given:

$$X \sim N(0, 1), \quad Y \sim N(0, 1)$$

Independent.

Using convolution:

$$Z = X + Y$$

Result:

$$\boxed{Z \sim N(0, 2)}$$

3.5 Exponential Case

If X, Y exponential with parameter λ:

$$f_Z(z) = \int_0^z \lambda e^{-\lambda x} \lambda e^{-\lambda(z-x)} dx$$

End of Lecture 11 Notes